

Chapter 5

Osseointegration

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Introduction

The fully healed site of the edentulous ridge (see Chapter 3) is most often covered by a masticatory mucosa that is about 2–3 mm thick. The masticatory mucosa is covered by a keratinized epithelium and includes a connective tissue rich in collagen fibers and fibroblasts that are firmly attached to the bone via the periosteum. The outer walls of the edentulous ridge, the cortical plates, are comprised of lamellar bone and enclose the cancellous bone that contains trabeculae of lamellar bone that are embedded in bone marrow. The bone marrow contains numerous vascular structures as well as adipocytes and pluripotent mesenchymal cells.

Different implant systems have been used to replace missing teeth, including subperiosteal implants, endosseous implants with fibrous encapsulation, and endosseous implants with direct bone contact (*osseointegrated*).

One definition of *osseointegration* [a term originally proposed by Brånemark *et al.* (1969)] was provided by Albrektsson *et al.* (1981) who suggested that this was “a direct functional and structural connection between living bone and the surface of a load carrying implant”. Another definition was provided by Zarb and Albrektsson (1991) who proposed that *osseointegration* was “a process whereby clinically asymptomatic rigid fixation of alloplastic materials is achieved and maintained in bone during functional loading”.

Schroeder *et al.* (1976, 1981, 1995) used the term “*functional ankylosis*” to describe the rigid fixation of the implant to the jaw bone, and stated that “new bone is laid down directly upon the implant surface, provided that the rules for atraumatic implant placement are followed and the implant exhibits primary stability”.

Thus, in order to acquire proper conditions for osseointegration (or functional ankylosis), the implant must exhibit proper initial fixation (primary stability) following installation in the recipient site. This initial or primary stability is the result of the contact relationship or friction that is established between mineralized bone (often the cortical bone) at the recipient site and the metal device.

Implant installation

Tissue injury

Basic rule: The less traumatic the surgical procedure and the smaller the tissue injury (the damage) in the recipient site during implant installation, the more expeditious is the process through which new bone is formed and laid down on the implant surface.

The various steps used at implant installation, such as (1) *incision* of the mucosa, often but not always followed by (2) the elevation of *mucosal flaps* and the separation of the periosteum from the cortical plates, (3) the preparation of the *canal* in the cortical and

spongy (cancellous) bone of the recipient site, and (4) the insertion of the titanium device (the implant) into this canal, bring to bear a series of mechanical insults and injury to both the mucosa and the bone tissue. The host responds to this injury with an inflammatory reaction, the main objective of which is to eliminate the damaged portions of the tissues and prepare the site for regeneration or repair. To the above described injury to the hard tissues must be added the effect of the so-called “press fit”, that is when the inserted implant is slightly wider than the canal prepared in the host bone. In such situations, (1) the mineralized bone tissue around the implant is compressed and exhibits a series of microfractures, (2) the blood vessels, particularly in the cortical portion, of the canal will collapse, (3) the nutrition to the bone in this portion is compromised, and (4) the affected tissues most often become non-vital.

The injury to the soft and hard tissues of the recipient site, however, also initiates the process of wound healing that ultimately ensures that (1) the implant becomes “ankylosed” with the bone, that is osseointegrated, and (2) a delicate mucosal attachment (see Chapter 4) is established and a soft tissue seal forms that protects the bone tissue from substances in the oral cavity.

Wound healing

The healing of the severed bone following implant installation is a complex process that apparently involves different events in different compartments of the surgical site.

In the *cortical bone compartment*, the non-vital mineralized tissue must first be removed (resorbed) before new bone can form. In the *spongy (cancellous) compartment* of the recipient site, on the other hand, the surgically inflicted damage (preparation of the canal and the installation of the implant) results mainly in soft tissue (marrow) injury that initially involves localized bleeding and clot (coagulum) formation. The coagulum is gradually resorbed and becomes replaced with granulation tissue; in-growth (from the walls of the prepared canal) of blood vessels, leukocytes, and mesenchymal cells. As a result of the continuous migration of mesenchymal cells from the surrounding marrow, the granulation tissue in turn is replaced with provisional connective tissue (provisional matrix) and eventually with osteoid. In the osteoid, deposition of hydroxyapatite crystals will occur in the collagen network around the newly formed vascular structures. Hereby, immature woven bone is formed (for detail see Chapter 3) and sequentially osseointegration occurs.

Cutting and non-cutting implants

In this chapter only screw-shaped implants made of c.p. titanium will be discussed. The design of the metal device and the installation protocol followed



Fig. 5-1 “Non-cutting” implant (solid screw: Straumann® Implant System). (Courtesy of Institut Straumann AG.)

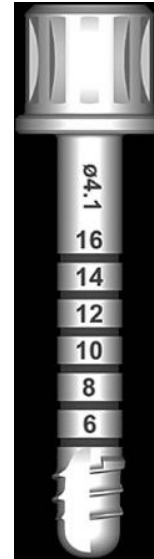


Fig. 5-2 Thread-tap (Straumann® Implant System) that is used to cut a track in the walls of the hard tissue canal. Following this preparation, the cavity in the host tissue and the implant are congruent. (Courtesy of Institut Straumann AG.)

may influence the speed of the process that leads to osseointegration.

“Non-cutting” implants (Fig. 5-1) require meticulous handling of the recipient site, including the preparation of a standardized track (thread) on the inside of the hard tissue canal. This track (thread) is prepared (precutting) using a thread-tap that is fitted with cutting edges (Fig. 5-2).

Figure 5.1 shows a “non-cutting” implant (e.g. solid screw, 4.1mm: Straumann® Implant System) that is designed as a cylinder with a rounded “apical” base. The diameter of the cylinder is 3.5mm. Pilot and twist drills of gradually increasing dimension are used to prepare the hard tissue canal of the recipient site to a final diameter of 3.5mm. On the surface of the cylinder, the implant is designed with a helix-shaped pitch that is 0.3mm high. The diameter of the entire screw-shaped device therefore becomes 4.1mm.

In sites with a high bone density, a thread-tap (Fig. 5-2) is used to cut a 4.1-mm wide helix-shaped track in the walls of the hard tissue canal. The implant and the cavity prepared in the hard tissues of the recipient site become congruent. When the implant is



Fig. 5-3 Ground section of a “non-cutting” implant and surrounding tissues obtained from a biopsy performed 24 hours after implant installation.

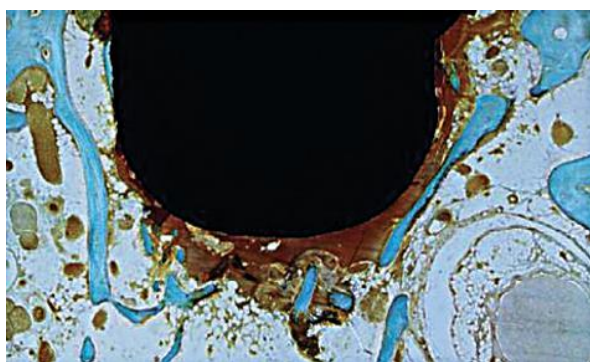


Fig. 5-4 Detail from the apical region of the implant described in Fig. 5-3. Note the presence of a coagulum in the bone marrow.

installed, the pitch on the device will capture and follow the helix-shaped track on the walls of the hard tissue canal and thereby guide the implant with a minimum of force into the preprepared position (Fig. 5-3).

Figure 5-3 illustrates a “non-cutting” Straumann® solid screw with surrounding tissues in a biopsy sampled 24 hours after implant installation. Proper initial fixation (stability) of the implant was obtained by the large contact area that was achieved between the metal screw and the buccal and lingual bone walls in the cortical compartment of the recipient site. During site preparation and placement of the implant, bone trabeculae in the spongy compartment of the site were obviously dislocated into the bone marrow. Blood vessels in the marrow compartment were severed, bleeding provoked, and a coagulum formed (Fig. 5-4).

After 16 weeks of healing (Fig. 5-5) the marginal portions of the “non-cutting” implant are surrounded by dense lamellar bone that is in direct contact with

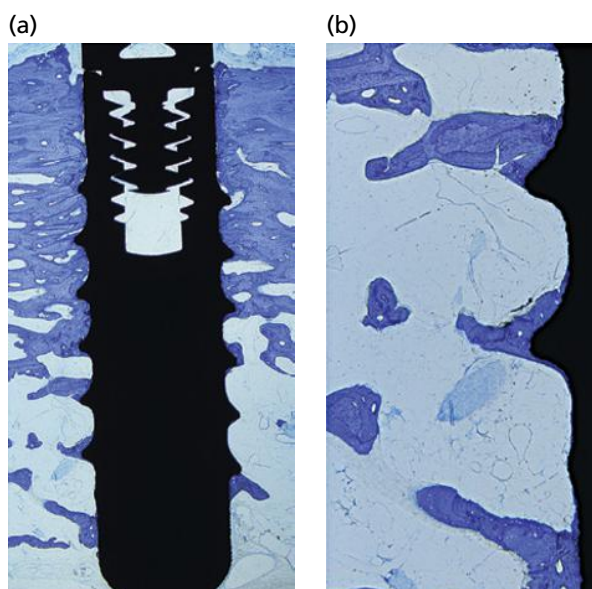


Fig. 5-5 (a) Ground section showing a “non-cutting” implant and surrounding bone after 16 weeks of healing. In the cortical portion of the recipient site, the bone density is high. (b) Detail of (a). In more apical areas, a thin coat of bone is present on the implant surface. Note also the presence of trabeculae of lamellar bone that extend from the implant into the bone marrow.

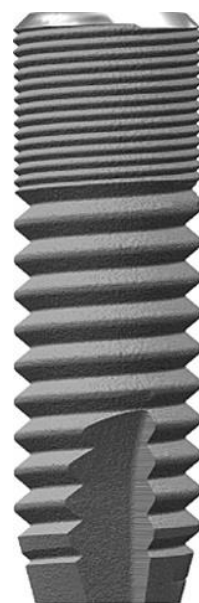


Fig. 5-6 “Cutting” implant (AstraTech® Implant System). Note the presence of cutting edges in the “apical” portion of the implant. During insertion this implant will cut a 0.3-mm wide chip from the lateral border of the canal prepared in the recipient site. (Courtesy of AstraTech.)

the rough surface of the metal device. Also, in the apical portion of the implant, a thin coat of mature bone can be seen to contact the implant surface and to separate the titanium screw from the bone marrow.

Cutting or self-tapping implants (e.g. AstraTech® Implant System, diameter 4.0mm) (Fig. 5-6) are designed with cutting edges placed in the “apical” portion of the screw-shaped device. The threads of the screw are prepared during manufacturing by cutting

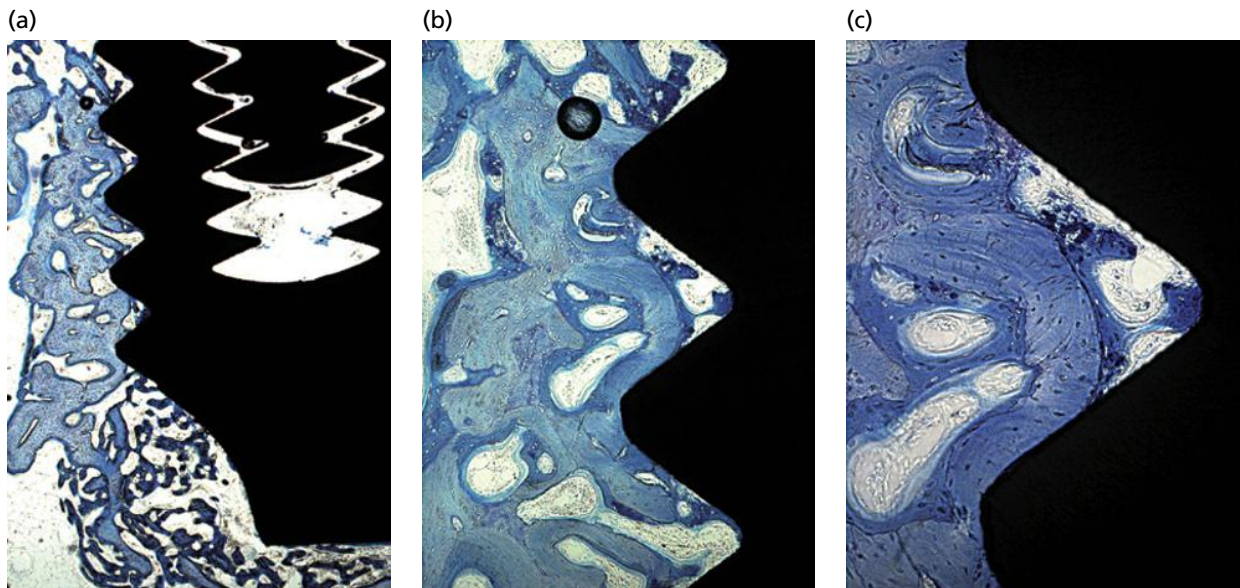


Fig. 5-7 (a) Ground section of an implant (AstraTech® Implant System) site from a biopsy sampled after 2 weeks of healing. In the apical area, large amounts of woven bone have formed. (b) Detail of (a). In the threaded region, newly formed bone can be seen to reach contact with the implant surface. (c) Higher magnification of (b). Newly formed bone extends from the old bone and reaches the titanium surface in the invagination between two consecutive “threads”.

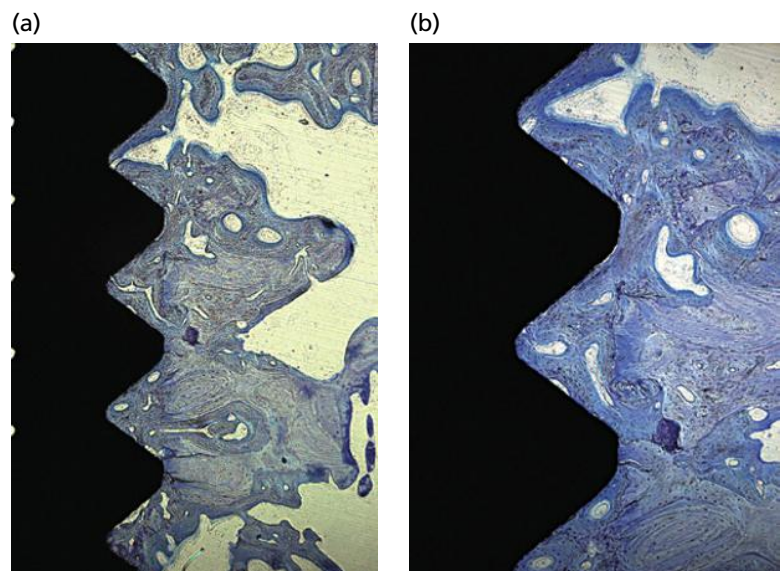


Fig. 5-8 Ground section of an implant site (AstraTech® self-tapping implant) from a biopsy specimen obtained after 6 weeks of healing. (a) In the marginal area, a continuous layer of bone covers most of the TiOblast® surface. (b) Higher magnification. Note the zone of newly formed (darker stained) bone that is in direct contact with the implant surface.

a continuous groove into the body of the titanium cylinder. When a self-tapping 4.0-mm wide implant is to be placed, the recipient site is first prepared with pilot and twist drills to establish a hard tissue canal that may have a final diameter of 3.7 mm. During insertion, the cutting edges in the “apical” portion of the implant create a 0.3-mm wide track in the walls of the canal and thereby establish the final 4.0 mm dimension. When the implant has reached its insertion depth, contact has been established between the outer portions of the threads and the mineralized bone in the cortical compartment (initial or primary fixation is thereby secured) and with the severed bone marrow tissue in the spongy (cancellous) bone compartment.

Figure 5-7 illustrates a recipient site with a self-tapping implant (AstraTech® implant). This implant is designed with a rough surface modification. The

biopsy was harvested 2 weeks after installation surgery. The outer portion of the thread is in contact with the “old” bone, while new bone formation is the dominant feature in the invaginations between the threads and in areas lateral to the “apical” portions of the implant. Thus, discrete areas of newly formed bone can be seen also in direct contact with the implant surface. In sections representing 6 weeks of healing (Fig. 5-8), it was observed that a continuous layer of newly formed bone covers most of the rough implant surface. This newly formed bone is also in contact with the old, mature bone that is present in the periphery of the recipient site. After 16 months of healing (Fig. 5-9), the bone tissue in the zone of osseointegration has remodeled and the entire hard tissue bed for the implant is comprised of lamellar bone including both concentric and interstitial lamella.

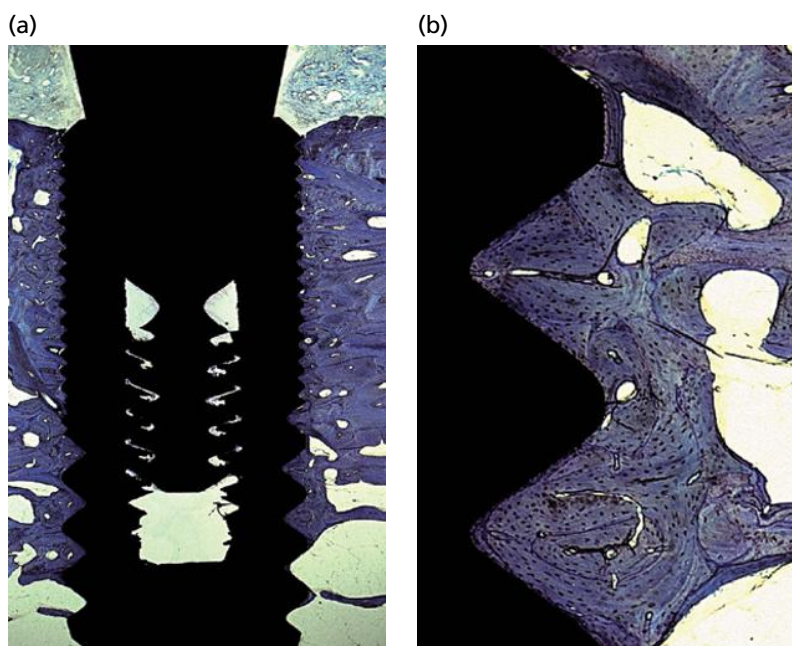


Fig. 5-9 Ground section of an implant site representing 16 months of healing. (a) The implant is surrounded by dense lamellar bone. (b) Higher magnification.

Process of osseointegration

De novo bone formation in the severed alveolar ridge following implant placement was studied in experiments in various experimental animal models (for review see Schroeder *et al.* 1995).

More recently, Berglundh *et al.* (2003) and Abrahamsson *et al.* (2004) described various steps involved in bone formation and osseointegration to implants placed in the mandible of dogs.

The device: Custom-made implants (made of c.p. titanium) in the shape of a solid screw and configured with a rough surface topography were utilized (Fig. 5-10). In the implant device, the distance between two consecutive profiles of the pitch (i.e. the threads in a vertical cross-section) was 1.25 mm. A 0.4-mm deep U-shaped circumferential trough had been prepared within the thread region during manufacturing (Fig. 5-11). The tip of the pitch was left untouched. Following the installation of the non-cutting device (Fig. 5-12), the pitch was engaged in the hard tissue walls prepared by the cutting/tapping device. This provided initial or primary fixation of the device. The void between the pitch and the body of the implant established a geometrically well-defined wound chamber (Fig. 5-13). Biopsies were performed to provide healing periods extending from 2 hours following implant insertion to 12 weeks of healing. The biopsy specimens were prepared for ground sectioning as well as for decalcified sectionings.

The wound chamber: Figure 5.13 illustrates a cross-section (ground section) of an implant with surrounding soft and hard tissues from a biopsy specimen sampled 2 hours after installation of the metal device. The peripheral portions of the pitch were in contact with the invaginations of the track prepared by the tap in the cortical bone. The wound



Fig. 5-10 Device used in the dog experiment. The implant is a modification of a solid screw (Straumann® Implant System). The distance between two consecutive threads is 1.25 mm. The depth of the trough is 0.4 mm. (Courtesy of AstraTech.)

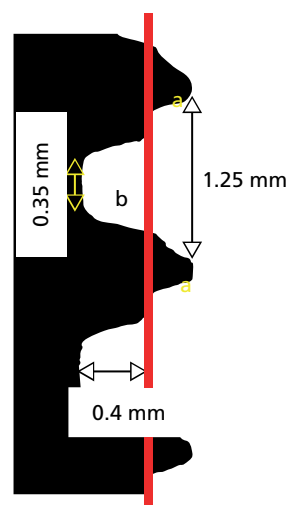


Fig. 5-11 The device. Schematic drawing illustrating the dimensions of the "wound chamber".

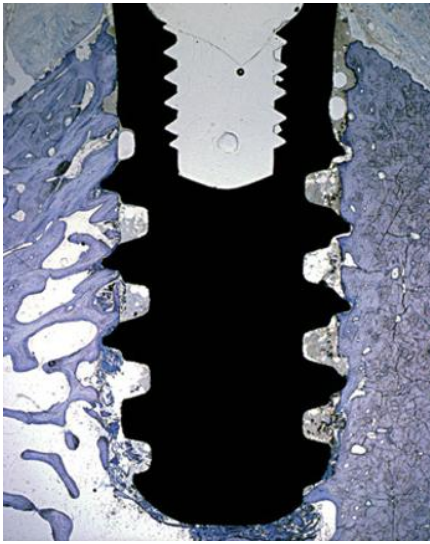


Fig. 5-12 Ground section showing the implant and adjacent tissues immediately after implant installation. The pitch region is engaged in the hard tissue walls. The void between two consecutive pitch profiles includes the wound chamber.

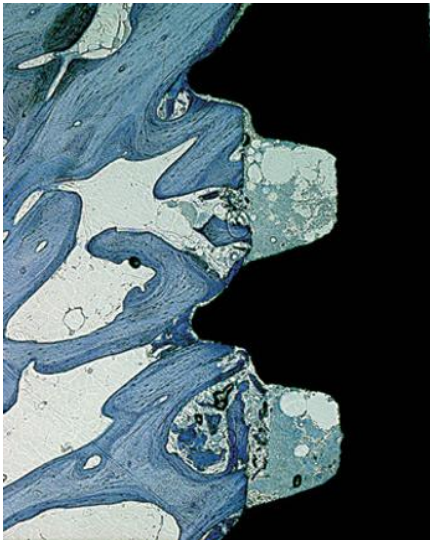


Fig. 5-13 Detail of Fig. 5-12. The wound chamber was filled with blood and a coagulum has formed.

chambers (Fig. 5-14a) were occupied with a blood clot in which erythrocytes, neutrophils, and monocytes/macrophages occurred in a network of fibrin (Fig. 5-14b). The leukocytes were apparently engaged in the wound cleansing process.

Fibroplasia: Figure 5-15a illustrates a device with surrounding tissues after 4 days of healing. The coagulum had in part been replaced with granulation tissue that contained numerous mesenchymal cells, matrix components, and newly formed vascular structures (angiogenesis) (Fig. 5-15b). A *provisional connective tissue (matrix)* had been established.

Bone modeling: After 1 week of healing, the provisional connective tissue in the wound chambers was rich in vascular structures and contained numerous mesenchymal cells (Fig 5-16a). The number of remaining inflammatory cells was relatively small. In several compartments of the chamber, a cell-rich immature bone (woven bone) was seen in the provisional connective tissue that surrounded the blood vessels. Woven bone formation occurred in the center of the chamber as well as in discrete locations that apparently were in direct contact with the surface of the titanium device (Fig. 5-16b). This was considered to represent the very first phase of osseointegration; contact between the implant surface and newly formed woven bone.

After 2 weeks of healing, woven bone formation appeared to be pronounced in all compartments, apical as well as lateral, surrounding the implant (Fig. 5-17a). Large areas of woven bone were found in the bone marrow regions “apical” of the implant. In the wound chamber, portions of the newly formed woven bone apparently extended from the old bone into the provisional connective tissue (Fig. 5-17b) and had in many regions reached the surface of the titanium device. At this interval, most of the implant surface was occupied by newly formed bone and a more comprehensive and mature osseointegration had been established (Fig. 5-17c). In the pitch regions, there were signs of ongoing new bone formation (Fig. 5-17d). Thus, areas

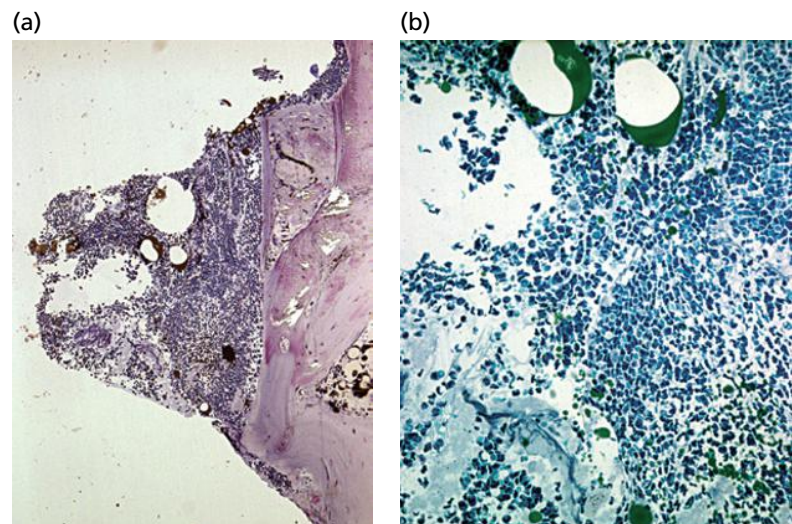


Fig. 5-14 Wound chamber 2 hours after implant installation. Decalcified sections. (a) The wound chamber is filled with blood. (b) Erythrocytes, neutrophils, and macrophages are trapped in a fibrin network.

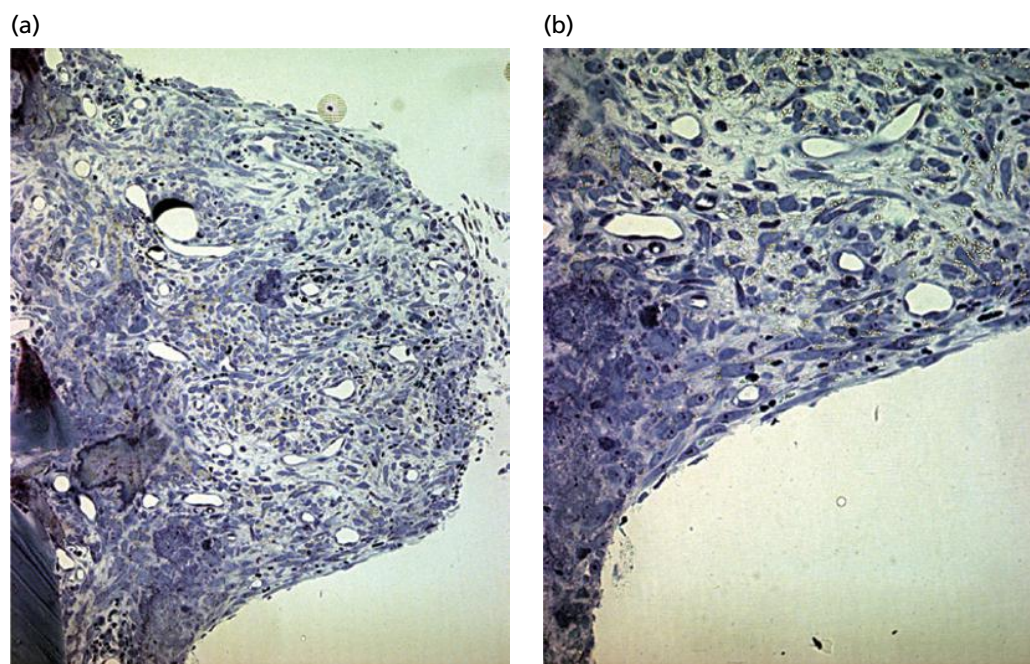


Fig. 5-15 Wound chamber after 4 days of healing (decalcified sections). (a) Most portions of the wound chamber are occupied by granulation tissue (fibroplasia). (b) In some areas of the chamber, provisional connective tissue (matrix) is present. This tissue includes large numbers of mesenchymal cells.

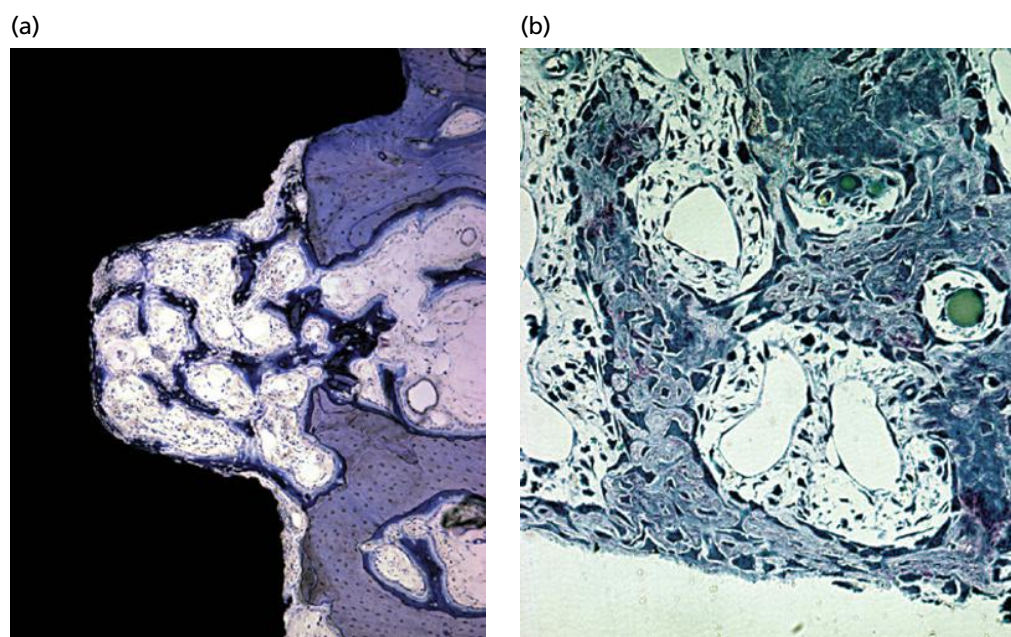


Fig. 5-16 (a) Ground section representing 1 week of healing. Note the presence of newly formed woven bone in the wound chamber. (b) Decalcified section. The woven bone is in direct contact with the implant surface.

of the recipient site located lateral to the device, that were in direct contact with the host bone immediately following installation surgery and provided initial fixation for the implant, had undergone resorption and were also involved in new bone formation after 2 weeks of healing.

At 4 weeks (Fig. 5-18a), the newly formed mineralized bone extended from the cut bone surface into the chamber and a continuous layer of cell-rich, woven bone covered most of the titanium wall of

the chamber. The central portion of the chamber was filled with a primary spongiosa (Fig. 5-18b), rich in vascular structures and a multitude of mesenchymal cells.

Remodeling: After 6–12 weeks of healing, most of the wound chambers were filled with mineralized bone (Fig. 5-19). Bone tissue, including primary and secondary osteons, could be seen in the newly formed tissue and in the mineralized bone that made contact with the implant surface. Bone marrow that contained

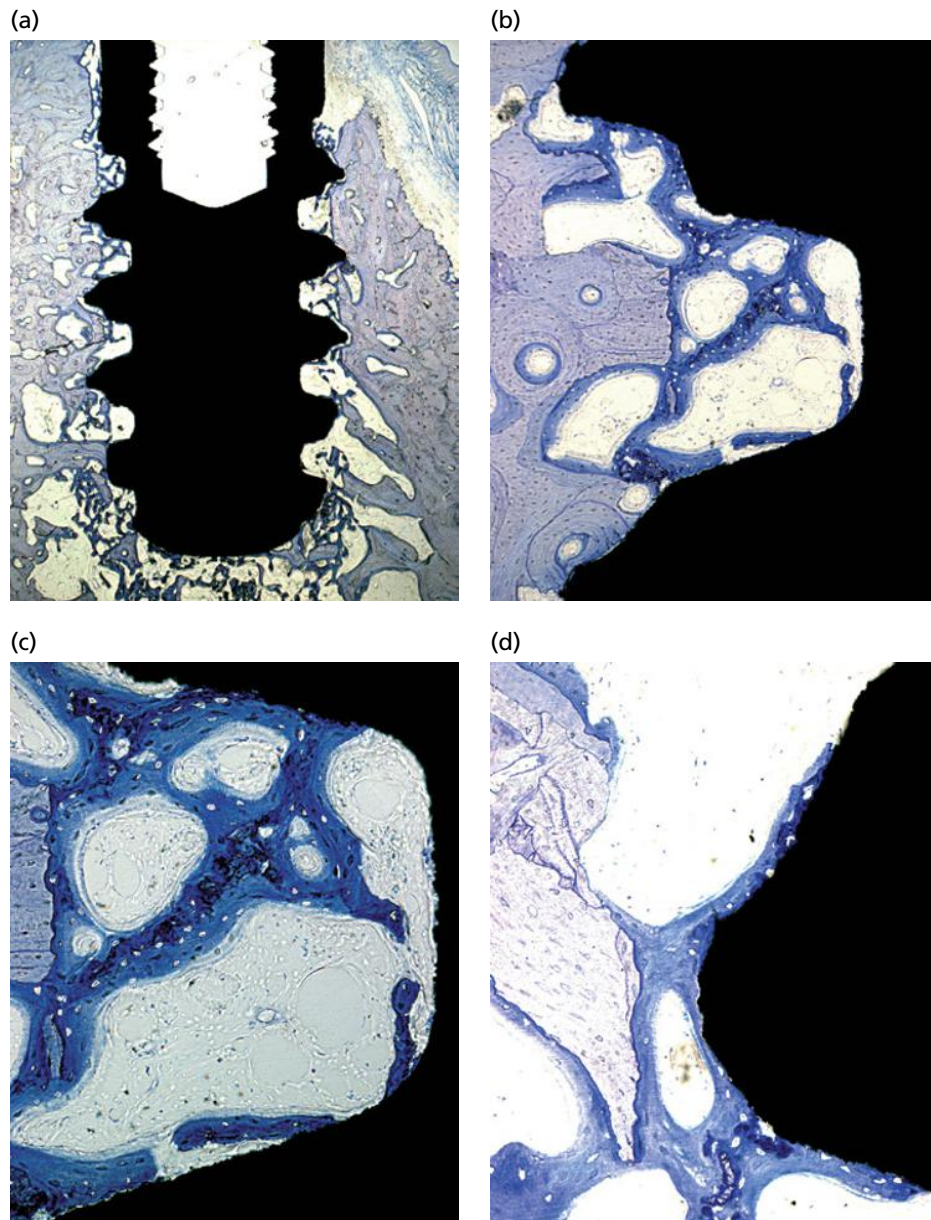


Fig. 5-17 Ground sections showing, in various magnifications, the tissues in the wound chamber after 2 weeks of healing. (a) Darker stained woven bone is observed in the apical area of the metal device. (b–d) Most portions of the implant surface are coated with bone.

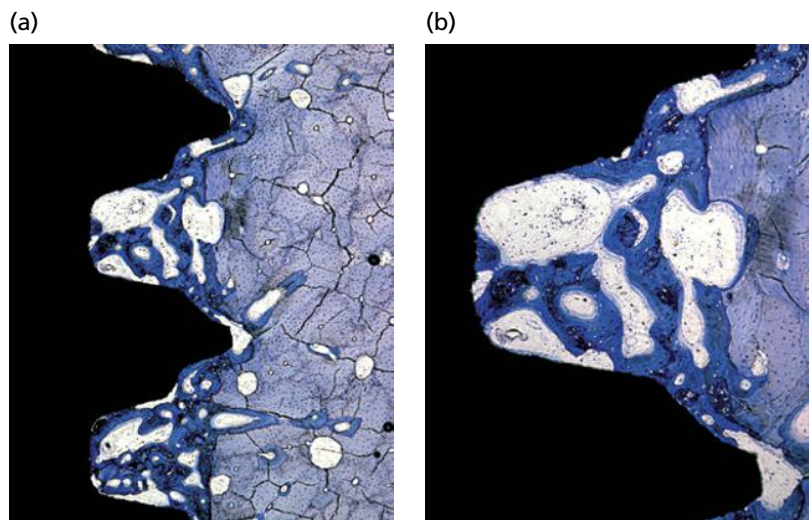


Fig. 5-18 Ground sections representing 4 weeks of healing. (a) Newly formed bone (dark blue) extends from the “old” bone into the wound chamber. (b) Appositional growth. Note the presence of primary osteons.

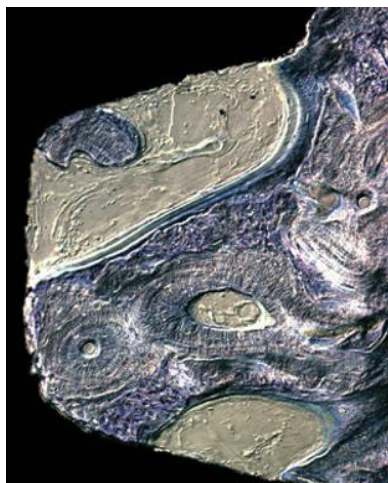


Fig. 5-19 Ground section representing 12 weeks of healing. The woven bone is being replaced with lamellar bone and marrow. Note the formation of secondary osteons.

blood vessels, adipocytes, and mesenchymal cells was observed to surround the trabeculae of mineralized bone.

Summary: The wound chambers were first occupied with a coagulum. With the in-growth of vessels and migration of leukocytes and mesenchymal cells, the coagulum was replaced with granulation tissue. The migration of mesenchymal cells continued and the granulation tissue was replaced with a provisional matrix, rich in vessels, mesenchymal cells, and fibers. The process of *fibroplasia* and angiogenesis had started. Formations of newly formed bone could be recognized already during the first week of healing; the newly formed woven bone projected from the lateral wall of the cut bony bed (appositional bone formation; distance osteogenesis) (Davies 1998), but *de novo* formation of new bone could also be seen on the implant surface, that is at a distance from the parent bone (contact osteogenesis) (Davies 1998). During subsequent weeks, the trabeculae of woven bone were replaced with mature bone, that is lamellar bone and marrow (bone remodeling).

Morphogenesis of osseointegration

A series of publications have described the process of osseointegration of titanium implants placed in human volunteers (Bosshardt *et al.* 2011; Donos *et al.* 2011; Ivanovski *et al.* 2011; Lang *et al.* 2011). In these studies, solid screw devices (Institute Straumann AG, Basel, Switzerland) with a moderately rough surface were placed in the retromolar region of the mandible and submerged healing conditions were established. Biopsies including the implant with surrounding tissues were retrieved with the use of a trephine drill after 1, 2, 4, and 6 weeks. The examination of the samples included histologic and morphometric measurements and particular attention was paid to tissue elements that were in direct contact with or close to the implant surface (the tissue–implant interface), for

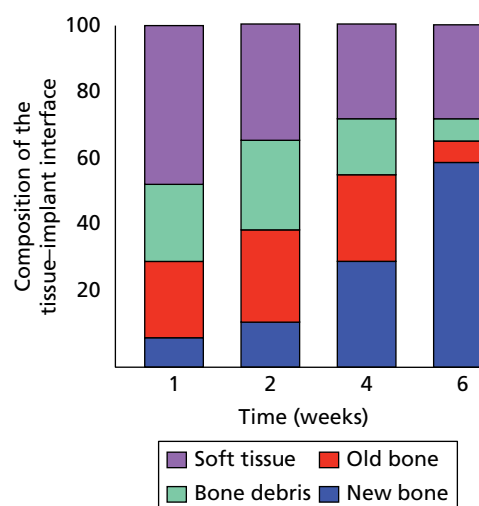


Fig. 5-20 Histogram showing the percentages of new bone, old bone, bone debris, and soft tissue in the “tissue–implant interface” after 1, 2, 4, and 6 weeks of healing. Note that the percentage of old bone, soft tissue, and bone debris that was present in the zone next to the implant surface decreased over time and that the amount of newly formed bone increased. There are reasons to suggest that (1) the contact between old bone and the implant established the initial “mechanical” stability of the titanium device while (2) the newly formed bone subsequently achieved osseointegration. (Courtesy of D.D. Bosshardt.)

example old bone, osteoid, newly formed bone, and non-mineralized mesenchymal soft tissue. In addition, at all examination intervals bone debris and solid bone particles were present in the wound lateral to the implant. Such constituents were obviously remnants of the drilling procedure used to prepare the hard tissue canal into which the implant was subsequently introduced.

Overall pattern of implant integration

Figure 5-20 describes the changes in the morphometric measurements in the tissue–implant interface region during the course of the study. After 1 week of healing, about 40% of the interface region was made up of soft tissue (granulation tissue, provisional connective tissue) and an additional 45% of bone debris and old bone. After 2 weeks, the amount of newly formed bone was still small, but the amount of soft tissue was markedly reduced. In the interval between 2 and 4 weeks, new bone formation was apparently pronounced in the interface zone. Thus, in this interval, newly formed bone increased from about 10% to about 30%, while the amount of hard tissue debris was markedly reduced. Also, in the period between 4 and 6 weeks, new bone formation was pronounced (from 30% to about 60%) and the diminution of old bone and bone debris markedly decreased. In other words, in humans the process of osseointegration appears to be most active in the interval between 2 and 6 weeks.

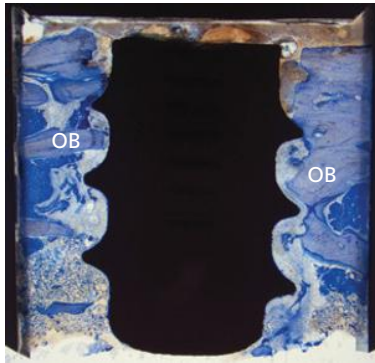


Fig. 5-21 Longitudinal ground section through a biopsy including a solid screw implant device. While compact old bone (OB) is found in contact with the coronal portion of the implant, the apical portion is comprised of less dense tissues and debris.

Summary: During the 6 weeks of healing that was monitored in this particular study in humans, it was observed that while the amount of old bone, bone debris, and soft tissue that initially occurred in close proximity to the implant gradually decreased, the amount of newly formed bone increased (Fig. 5-20). This pattern of healing that eventually resulted in osseointegration is in close agreement with the results obtained from the animal experiments reported earlier in this chapter.

Biopsy sample observations

Early wound

An implant with surrounding tissues sampled in the early phase after the surgical installation of the device is shown in Fig. 5-21. Note the presence of old bone, particularly in the cortical (marginal) region of the site. This old compact bone appeared to be in direct contact with the implant and obviously facilitated the initial mechanical stability of the device. Note also that more apical portions of the implant were surrounded by non-mineralized tissue, bone debris, and bone particles.

Healing process

After 1 week of healing, substantial amounts of old bone occupied the marginal portion of the surgically prepared site. This bone tissue appeared to be in close contact with the implant device (Fig. 5-22). As stated above, this close fit between the remaining old bone and the titanium device was most likely a prerequisite for initial implant stability and of importance in establishing optimal healing conditions in the hard tissue wound. At this early interval, newly formed bone occurred on the surface of old bone tissue (Fig. 5-23), while areas of bone resorption could be identified in adjacent regions of the tissue wound. In other words, phenomena such as hard tissue apposition and resorption characterized the healing process in this early phase.

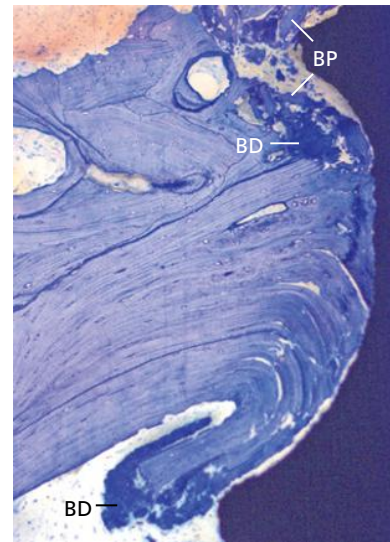


Fig. 5-22 Compact bone in direct contact with the implant surface in the coronal portion after 1 week of healing. Note the presence of bone particles (BP) and bone debris (BD) of varying size close to the implant surface. (Courtesy of D.D. Bosshardt.)

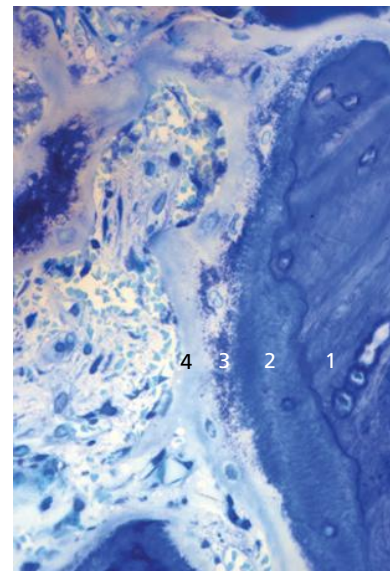


Fig. 5-23 Initial stage of bone apposition onto the surface of old bone occurring at a distance from the implant surface after 1 week of healing. 1, old bone; 2, new mineralized bone matrix; 3, mineralization foci at the mineralization front; 4, osteoid lined by osteoblasts.

Bone debris, bone particles, soft mesenchymal tissue as well as thin layers of osteoid tissue were also frequently found on or close to the implant surface (Figs. 5-24, 5-25).

At the 2-week interval, remnants of old bone apparently still remained in the marginal portion of the implant site. Areas of hard tissue resorption (Howship's lacunae; Fig. 5-26) could be found immediately adjacent to as well as at a distance from the implant. In addition, minute areas of newly formed bone occurred on or immediately lateral to the surface of the implant device. This formation of woven bone was the first sign of what may be called

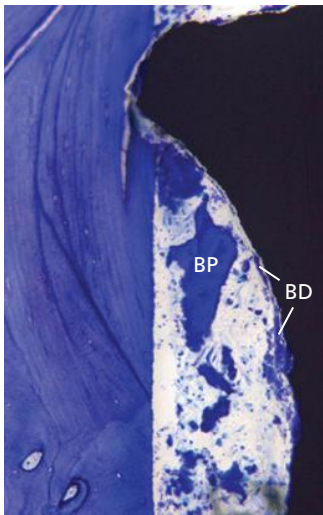


Fig. 5-24 After 1 week of healing, a considerable amount of bone debris (BD) and larger bone particles (BP) are present in the gap between the implant surface and the cut bone bed. (Courtesy of D.D. Bosshardt.)

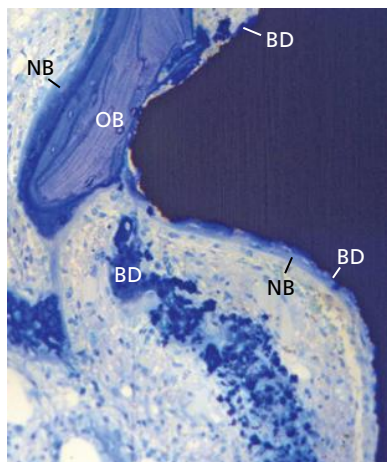


Fig. 5-25 After a healing period of 1 week, old bone (OB) is still in contact with the pitch of the implant thread. Newly formed bone (NB) is present (1) on the ledges of old bone and (2) on the implant surface. Bone debris (BD) is found adhering to the implant surface, but is also embedded in the adjacent mesenchymal soft tissue. The newly formed bone mainly consists of a partly mineralized osteoid lined by osteoblasts.

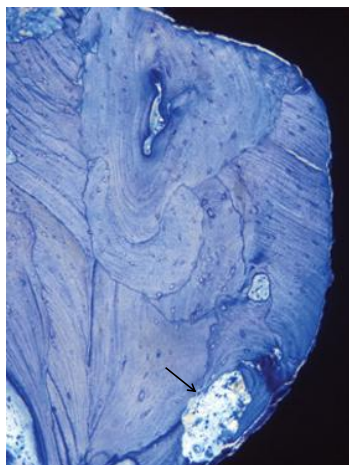


Fig. 5-26 Area of compact old bone in contact with the most coronal portion of the implant after a healing period of 2 weeks. Note the presence of bone resorption at the bottom of the micrograph (arrow).

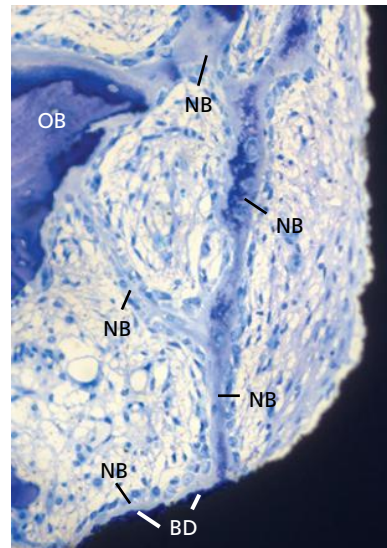


Fig. 5-27 Site characterized by active tissue modeling, in other words woven bone formation. The newly formed trabeculae of woven bone extend from old bone into the provisional connective tissue. (OB, old bone; NB, new bone; BD, bone debris.)

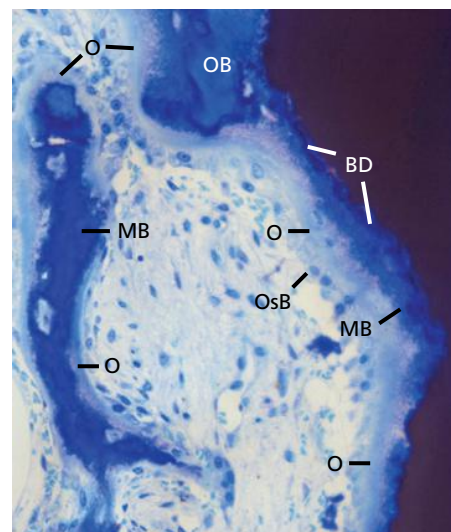


Fig. 5-28 Micrograph showing the implant-tissue interface of an implant site after 2 weeks of healing. The area is filled with a provisional connective tissue matrix and zones of new bone are discernible as well as osteoid tissue on the implant surface. Note the presence of bone debris (BD) on the surface of the implant. Tissue elements, including mineralized matrix of immature bone (MB) as well as osteoid tissue (O) and old bone (OB), are in contact with the implant surface. OsB, border between osteoid and connective tissue. (Courtesy of D.D. Bosshardt.)

osseointegration (Figs. 5-27, 5-28). Furthermore, at this interval, tiny ledges of newly formed woven bone apparently connected old bone to the titanium screw device (Fig. 5-28).

At the 4-week interval, the healing process features of modeling and remodeling were pronounced. Thus, in some areas close to the implant surface resorptive processes were discernible, while in adjacent areas woven bone had formed (Fig. 5-29).

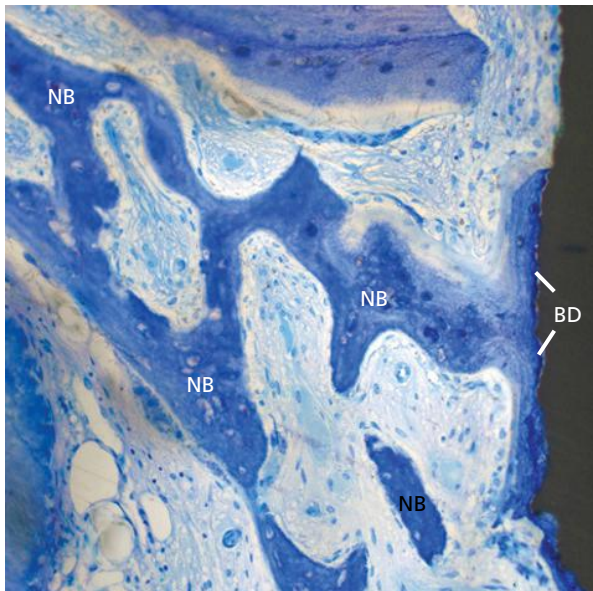


Fig. 5-29 Micrograph showing the implant–tissue interface and the peri-implant tissues of an implant after 4 weeks of healing. The newly formed bone (NB) forms a tiny trabecular network connecting the surface of the old bone with the implant surface. Deposition of new bone on the implant surface was associated with the presence of bone debris (BD). (Courtesy of D.D. Bosshardt.)

At the 6-week interval, large amounts of newly formed woven bone (Fig. 5-30), but also lamellar bone and marrow, were present in close proximity to the implant device. This kind of newly formed hard tissue was apparently part of a more stable “bone–implant contact”, in other words osseointegration.

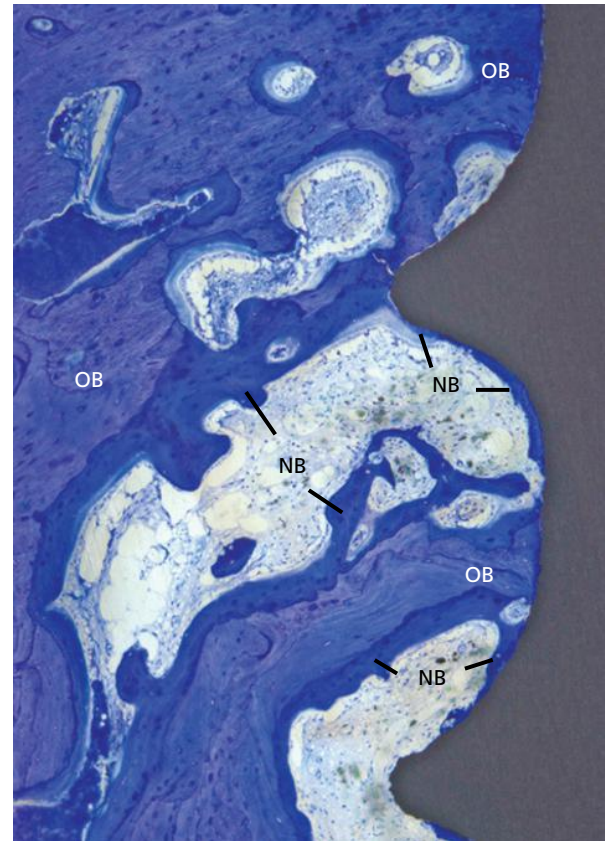


Fig. 5-30 Microphotograph showing the implant–tissue interface after 4–6 weeks of healing. New bone (NB) is found on the surface of old bone (OB) and on the implant surface. (Courtesy of D.D. Bosshardt.)

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